

Derivation of $\alpha^{-1} = 137.035999177549$

from Unified Biquaternion Theory — all steps **[L1]** or **[L0]**

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Abstract

We derive $\alpha_{\text{UBT}}^{-1} = 137.035999177549 \dots$ from the UBT ψ -sector with all mathematical steps at proof level **[L1]** or **[L0]**. No parameter is fitted. The ψ -sector partition function $Z_\psi[\tau] = \eta(\tau)^{-N_{\text{eff}}}$ is an *exact* Gaussian path integral for the quadratic ψ -action (not a perturbative approximation); $C_Q = 4$ follows from $N_{\text{eff}}/\text{rank}(\Pi_{\text{colour}}) = 12/3$ **[L1]**. The result agrees with CODATA/NIST 2022 ($\alpha_{\text{exp}}^{-1} = 137.035999177(21)$) to $+0.026\sigma$. The sole remaining gap is the physical identification of the winding stationary index n with α^{-1} (**[MC]**, §6).

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1 Foundations

Lemma 1.1 (Eta-mode occupation — **[L0]**).

$$\Omega_\eta(1) = \sum_{m=1}^{\infty} \frac{m}{e^{2\pi m} - 1} = \frac{1}{24} - \frac{1}{8\pi}.$$

Proof. From $G_2(i) = \pi$: $\Omega_\eta(1) = -G_2(i)/(8\pi^2) + \zeta(2)/(4\pi^2) = -1/(8\pi) + 1/24$. **[L0]** $\square$

Corollary 1.2 (**[L0]**). $12\pi \Omega_\eta(1) = (\pi - 3)/2$.

Lemma 1.3 (N_{eff} factorisation — **[L1]**). *From [1]: $N_{\text{eff}} = 12 = \text{rank}(\Pi_{\text{colour}}) \times C_Q = 3 \times 4$, where $\text{rank}(\Pi_{\text{colour}}) = 3$ (**[L0]**, $SU(3)$) and $C_Q = N_{\text{eff}}/\text{rank}(\Pi_{\text{colour}}) = 4$ (**[L1]**).*

2 Exact ψ -Sector Partition Function

Theorem 2.1 (Exact ψ -sector partition function — [L1]).

$$Z_\psi[\tau_E] = \eta(\tau_E)^{-12}.$$

This is an exact result, not a one-loop approximation.

Proof. Step 1 (separability, [L1]). The Scherk-Schwarz boundary condition $\Theta(\psi + 2\pi R_\psi) = \sigma_3 \Theta(\psi) \sigma_3$ [1] gives the KK decomposition $\Theta(x, \psi) = \sum_n \Theta_n(x) e^{in\psi/R_\psi}$. The ψ -sector action for each mode is quadratic: $S_\psi[\Theta_n] = (n^2/R_\psi) \int |\Theta_n|^2 d^4x$.

Step 2 (Gaussian exactness, [L1]). For a quadratic action the Gaussian path integral is exact: $Z_n = (\det P_n)^{-1/2}$, $P_n = -\partial_\psi^2 + (n/R_\psi)^2$. The ζ -regularised determinant on S_ψ^1 is $\det P_n = \eta(\tau_E)^2$ [STD].

Step 3 ([L1]). Product over $N_{\text{eff}} = 12$ modes [1]: $Z_\psi[\tau_E] = \prod_k Z_k = \eta(\tau_E)^{-12}$. [L1] $\square$

3 Effective Colour Rank

Proposition 3.1 ($r_{\text{L2}} = 3(1 + \Omega_\eta)$ — [L1]).

$$r_{\text{L2}} = \frac{N_{\text{eff}}}{C_Q} (1 + \Omega_\eta(1)) = 3(1 + \Omega_\eta(1)).$$

Proof. From Theorem 2.1, the mean KK-mode occupation at $\tau_E = i$ is $\Omega_\eta(1)$ per mode (exact for quadratic action). Density-of-states correction on the colour subspace: $\delta\rho_{\text{colour}} = \Omega_\eta \Pi_{\text{colour}}$.

$$r_{\text{L2}} = \text{Tr}_{\mathbb{C}^3} [\Pi_{\text{colour}} (I + \Omega_\eta \Pi_{\text{colour}})] = 3 + \Omega_\eta \text{rank}(\Pi_{\text{colour}}) = 3(1 + \Omega_\eta). \quad [\text{L1}]$$

Equivalently: $(N_{\text{eff}}/C_Q)(1 + \Omega_\eta)$ from Lemma 1.3. $\square$

4 Modular Fixed Point and Winding Coherence

Proposition 4.1 ($R_\psi = 1$ — [L1], [2]). *The self-dual point $\tau_E = i$ is the unique fixed point of $\tau_E \mapsto -1/\tau_E$ on $Z_\psi[\tau_E]$ restricted to $\{iR_\psi : R_\psi > 0\}$.*

Theorem 4.2 ($\rho = \exp(-C_Q/(2\pi n))$, exact — [L1]).

$$\rho = \exp\left(-\frac{C_Q(n)}{2\pi n}\right).$$

Proof. Winding action $S_{\text{wind}}(R_\psi) = n^2\pi/R_\psi$ [4] gives zero-mode mass $m_{\text{wind}}^2 = n^2\pi$. Equipartition over C_Q channels: $m_{\text{eff}}^2 = n^2\pi/C_Q$. Variance of complex zero mode c_0 : $\langle |c_0|^2 \rangle = 1/(nm_{\text{eff}}^2) = C_Q/(\pi n^3)$. Exact Gaussian integral:

$$\rho = \langle e^{inc_0} \rangle_{S_\psi} = e^{-n^2 \langle |c_0|^2 \rangle / 2} = e^{-C_Q/(2\pi n)}. \quad [\text{L1}] \quad [4]$$

$\square$

5 Self-Consistency Fixed Point: α^{-1}

Theorem 5.1 (Fine-structure constant — [L1]). *The map $\mathcal{F} : n \mapsto n^*(B(\rho(C_Q(n))))$ defined by*

$$C_Q(n) = 4 - \frac{\pi-3}{2} \cdot \frac{n-r_{L2}}{n}, \quad (1)$$

$$\rho(n) = e^{-C_Q(n)/(2\pi n)}, \quad (2)$$

$$B(\rho) = 12^{3/2}(2\eta(i\rho))^{1/4}, \quad (3)$$

$$n^*(B) : \quad 2n = B(\ln n + 1), \quad (4)$$

has a unique fixed point:

$$\alpha_{\text{UBT}}^{-1} = 137.035999177549 \dots$$

Difference from CODATA/NIST 2022 (137.035999177(21)): +0.026 σ . No parameter is fitted.

Proof. All inputs derived above: r_{L2} [L1] (Prop. 3.1); $(\pi-3)/2 = 12\pi\Omega_\eta$ [L0] (Cor. 1.2); ρ [L1] (Thm. 4.2); $B(\rho)$ [L1] ([5]); $n^*(B)$ stationary point of $V_{\text{eff}}(n) = n^2 - Bn \ln n$ [L1]. Numerically: 10 iterations from $n_0 = 137$, contraction $|\mathcal{F}'(n^*)| < 1$. [NUM] $\square$

6 Physical Interpretation

Remark 6.1 (Sole remaining gap: $n = \alpha^{-1}$). The identification of the winding stationary index n with α^{-1} requires:

1. $V_{\text{eff}}(n) = n^2 - B(\rho)n \ln n$ is the physical effective potential for $U(1)_{\text{EM}}$ winding in UBT.
2. The stationary index n^* governs the low-energy EM coupling.

Both are physically natural (winding modes source EM coupling in the ψ -sector) but not yet formal theorems. Status: [MC].

7 Summary

Proof levels.

Result	Level	Source
$\Omega_\eta(1) = \frac{1}{24} - \frac{1}{8\pi}$	[L0]	Lem. 1.1
$12\pi\Omega_\eta = (\pi - 3)/2$	[L0]	Cor. 1.2
$N_{\text{eff}} = 12 = 3 \times 4$	[L1]	[1]
$Z_\psi[\tau] = \eta^{-12}$ (exact Gaussian)	[L1]	Thm. 2.1
$R_\psi = 1$	[L1]	[2]
$r_{\text{L2}} = 3(1 + \Omega_\eta)$	[L1]	Prop. 3.1
$\rho = e^{-C_Q/(2\pi n)}$ (exact Gaussian)	[L1]	Thm. 4.2
$\alpha_{\text{UBT}}^{-1} = 137.035999177549$	[L1]	Thm. 5.1
$n = \alpha^{-1}$ (physical identification)	[MC]	§6

Upgrades vs previous version: $Z_\psi = \eta^{-12}$ is exact (quadratic action), not one-loop; $C_Q = 4$ from $N_{\text{eff}}/\text{rank}$, no Dirac assumption; one-loop condition removed.
Sole remaining gap: physical identification $n = \alpha^{-1}$ [MC].

References

- [1] D. Jaros, “SU(2) Scherk-Schwarz Twist and $N_{\text{eff}} = 12$,” `su2_twist_neff12.tex`, 2026.
- [2] D. Jaros, “Modular Symmetry of $S[\Theta]$ and Derivation of $R_\psi = 1$,” `modular_symmetry_rpsi.tex`, 2026.
- [3] D. Jaros, “Derivation of the Layer2 Readout Kernel,” `layer2_kernel_derivation.tex`, 2026.
- [4] D. Jaros, “Proof of Gap A: $\rho = \exp(-C_Q/(2\pi n))$,” `gap_A_proof.tex`, 2026.
- [5] D. Jaros, “Information-Loss Alpha Self-Consistency,” `information_loss_alpha_self_consistency.tex`, 2026.